

Mutually exclusive events



- 1 Yes
- 2 No
- 3 No
- 4 Selecting a 5 and selecting a queen.
- 5 Rolling a 3 and rolling a multiple of 3.
- 6 C, D
- 7 $\frac{3}{4}$
- 8 1
- 9 0.36
- 10 a $\frac{26}{52}$ b $\frac{8}{52}$ c $\frac{2}{52}$

Non-mutually exclusive events

- 11 $P(A \cup B) = \frac{1}{3}$
- 12 $P(A \cap B) = \frac{2}{5}$
- 13 $P(B) = \frac{1}{5}$
- 14 $P(A \text{ or } B) = 0.95$
- 15 4



16 $P(A \text{ and } B) = \frac{1}{12}$

17 a $\frac{26}{52}$ b $\frac{4}{13}$ c $\frac{1}{26}$ d $\frac{8}{13}$

18 a $\frac{1}{6}$ b $\frac{3}{6}$ c $\frac{4}{6}$ d $\frac{5}{6}$
e $\frac{1}{6}$ f $\frac{5}{6}$ g $\frac{5}{6}$ h $\frac{2}{6}$
i $\frac{1}{6}$ j $\frac{3}{6}$

19 a $\frac{3}{20}$ b $\frac{3}{4}$

20 a $\frac{30}{67}$ b $\frac{32}{67}$ c $\frac{37}{67}$ d $\frac{35}{67}$
e $\frac{19}{67}$

21 a i 7 ii 8 iii 5 iv 3 b $\frac{15}{23}$
c $\frac{5}{23}$

22 a b i $\frac{16}{43}$ ii $\frac{115}{129}$ iii $\frac{14}{129}$



